The Learnability of Model-Theoretic Interpretation Functions in Artificial Neural Networks

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https://github.com/abrsvn/neural_model_theoretic_interpretation_ACL_2026

Central Question

Can NNs learn model-theoretic interpretation functions that generalize systematically to novel sentences?

The Systematicity Challenge

Fodor & Pylyshyn (1988) argued: Human cognition is **systematic** (understanding "cat on mat" implies understanding "dog on mat"); Systematicity requires compositional, symbolic representations; NNs systematic only by implementing a symbol system

Frank et al. (2009): Networks can exhibit systematicity **without** implementing symbols; Systematicity arises from **structure in the world**, encoded in analogical situation-vector representations; Situation vectors are NOT compositional/symbolic—no parts represent concepts

We extend this work with:

- Pool MCMC sampling (better joint-distribution estimates)
- Entity vectors (semantic type e extension)
- Modern architectures (GRU, LSTM, Transformers)
- Principled competing-event generation and an extended eval set (~ 350 vs. ~ 80 test sentences)

The Microworld

17 entities in 6 categories: **People** (3): charlie, heidi, sophia; **Games** (3): chess, hide&seek, soccer; **Toys** (3): puzzle, ball, doll; **Locations** (4): bathroom, bedroom, playground, street; **Play-manners** (2): well, badly; **Win-manners** (2): with_ease, with_difficulty

44 atomic propositions: play(person, game): 9; play(person, toy): 9; win(person), lose(person): 6; location(person, place): 12; play_manner(person, manner): 6; win_manner(manner): 2

167 hard constraints across multiple categories: Game rules: one game at a time, chess=2 players & soccer=2–3; Location rules: everyone has exactly one location, chess players co-located; Winning rules: can't both win and lose, only one winner; Toy rules: puzzle=1 player only, soccer requires ball

Situation space: $2^{44} \approx 17.6$ trillion $\rightarrow$ **28,676** valid situations

The Microlanguage

40 words in a feature-based CFG: Persons, games, toys, locations; Verbs: plays, wins, beats, loses, is, played, won, lost; Modifiers: well, badly, inside, outside, with ease/difficulty; Prep.s: with, to, at, in, by

4 synonym pairs: charlie=boy, soccer=football, puzzle=jigsaw, bathroom=shower

Total: 13,556 sentences (only some consistent with hard constraints)

Example translations: 'charlie plays chess' $\rightarrow$ play(charlie, chess); 'girl plays chess' $\rightarrow$ play(heidi, chess) $\vee$ play(sophia, chess); 'charlie loses to sophia' $\rightarrow$ win(sophia) $\wedge$ lose(charlie)

References. [1] Frank, Haselager & van Rooij (2009). Connectionist semantic systematicity. *Cognition* 110. [2] Fodor & Pylyshyn (1988). Connectionism and cognitive architecture. *Cognition*, 28. [3] Brasoveanu (2026). Distributed Situation Models: Probabilistic Representations of Truth-Conditions. *CogSci 2026*. [4] Hochreiter & Schmidhuber (1997). Long short-term memory. *Neural Computation* 9. [5] Cho et al. (2014). Learning Phrase Representations using RNN Encoder–Decoder for SMT. *EMNLP*. [6] Vaswani et al. (2017). Attention is all you need. *NeurIPS 30*. [7] Su et al. (2021). RoFormer: Enhanced transformer with rotary position embedding.

Our Extensions to Frank et al. (2009)

I. Pool MCMC Sampling

Problem: Frank et al. sampled propositions *sequentially*, evaluating soft constraints on partial information. This underestimates correlations.

Our solution (Brasoveanu 2026):

1. Pre-enumerate all 28,676 valid situations
2. MCMC over pool with Metropolis sampling
3. Acceptance based on **full-context** soft constraint probability
4. 1M raw samples $\rightarrow$ 250k thinned (factor 4)

Result: Correct joint distribution respecting both hard AND soft constraints.

II. Entity Vectors

Motivation: Formal semantics has two fundamental types—truth values (type t) AND entities (type e). Frank et al. encoded only truth values (want: entities w/o implicitly symbolic architecture).

Implementation:

1. Enhance MCMC samples: 44-dim $\rightarrow$ 61-dim (+ 17 entity indicators)
2. Joint competitive layer training on 61-dim samples
3. Extract: 150×44 props + 150×17 entities

Training targets: Without entities: 150-dim; With entities: 300-dim (concatenated)

III. Modern Architectures

Experiment 1 capacity-matched at $\approx$66k parameters:

Arch	Hidden	Layers	Heads	Params
SRN	178	1	—	66,010
GRU	90	1	—	66,210
LSTM	80	1	—	66,950
Attn AbsPE	48	2	4	65,670
Attn RoPE	48	2	4	65,670

AbsPE: Sinusoidal positional encoding (Vaswani et al. 2017)
RoPE: Rotary positional encoding (Su et al. 2021)

Follow-up Experiment 2: higher-capacity attention (H80) tests whether parameter count per input token closes the gap (does not).

IV. Principled Generation of Competing Events

Problem: Frank et al. used hardcoded rules per test group.

Our approach: Universal two-step competing event generation

Minimal difference for candidate generation: Vary exactly one event slot (agent, theme, location, manner) at a time.

Step 1 (Validity): Candidate + constraints $\rightarrow$ SAT
Step 2 (Competition): Candidate + described + constraints $\rightarrow$ UNSAT

Situation Vectors & Evaluation

Competitive Neural Model (Brasoveanu 2026, Frank et al. 2009):

- Input: 250k thinned MCMC samples ($44+17=61$-dimensional)
- 150 competitive units, L1 distance, winner-take-all
- Output: 150×61 weight matrix $\mathbf{W} \in [0,1]^{150\times 61}$. Meanings for atomic props p: first 44 columns $\mathbf{W}_{\cdot p} = [\![p]\!] \in [0,1]^{150}$ ($p = 1 \dots 44$). Entity-presence vectors: last 17 columns of $\mathbf{W}$.

Probability estimation (for any propositions p, q; $i = 1 \dots 150$):

$$P(p) \approx \frac{\Sigma_i [\![p]\!]_i}{150} \qquad P(p \wedge q) \approx \frac{\Sigma_i [\![p]\!]_i \cdot [\![q]\!]_i}{150} \qquad P(p|q) = \frac{P(p \wedge q)}{P(q)}$$

Fuzzy logic operations (truth conditions for non-atomic props):

$$[\![\neg p]\!]_i = 1 - [\![p]\!]_i \qquad [\![p \wedge q]\!]_i = [\![p]\!]_i \cdot [\![q]\!]_i$$
$$[\![p \vee q]\!]_i = [\![p]\!]_i + [\![q]\!]_i - [\![p]\!]_i \cdot [\![q]\!]_i$$

Comprehension Score (Frank et al.): attained fraction of max possible belief change in actual target proposition a given model output z.

$$\text{Comprehension}(a|z) = \begin{cases} \frac{P(a|z)-P(a)}{1-P(a)} & \text{if } P(a|z) > P(a) \\ \frac{P(a|z)-P(a)}{P(a)} & \text{otherwise} \end{cases}$$

Described vs. Competing Events:
Example: "charlie beats heidi" $\Rightarrow$ win(charlie) $\wedge$ lose(heidi)

- **Described**: win(charlie), lose(heidi)—should be **positive**
- **Competing**: win(heidi), lose(charlie)—should be **negative**

Systematicity = **Advantage** = score(described) − score(competing)
$\Rightarrow$ model correctly derives specific target truth conditions

Train/Test Design

Complementary splits: What's held out in Split 1 is trained in Split 2 (and vice versa)—ensuring results don't depend on which particular sentences are excluded (C=charlie, H=heidi, S=sophia).

Test Group	Split 1 Held Out	Split 2 Held Out	Truth Cond in Train
Word (easiest)	C+soccer	boy+football	yes
Sentence	C {beats/loses to} H …	C {beats/loses to} S …	yes
Complex Event	chess+outside …	chess+inside …	no
Basic (hardest)	C+doll …	C+ball …	no

Four test groups of increasing difficulty:

- **Word**: Novel word combinations (synonym substitution), but target truth cond seen in training
- **Sentence**: Novel person pairs in "beats"/"loses to"—can model learn argument alternations?
- **Complex**: Novel game+location conjunctions, target truth cond **not** seen in training (only truth cond for individual conjuncts seen)
- **Basic**: Novel person+toy combinations (hardest), target truth cond **not** seen in training

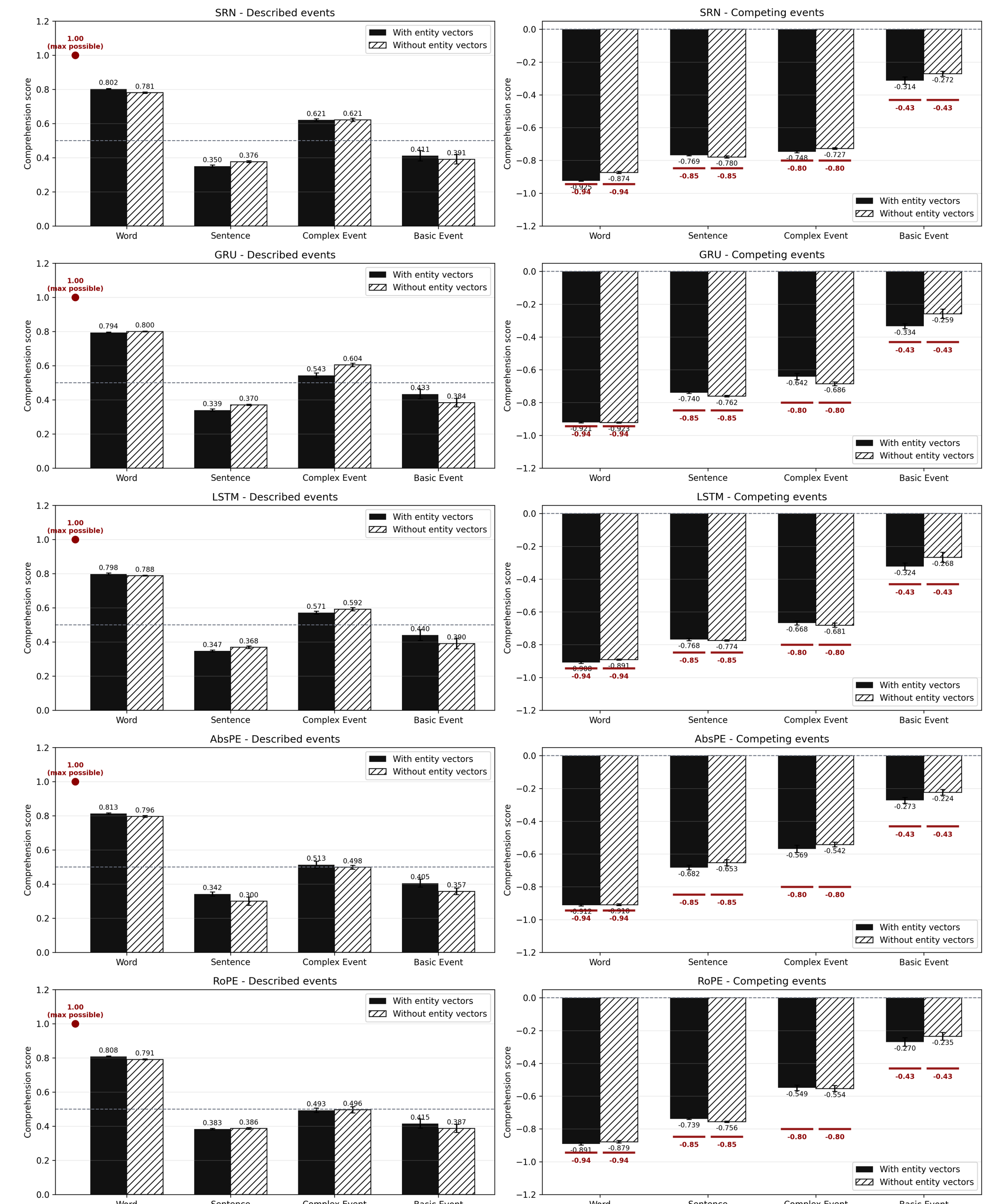
Grammar generates 13,556 sentences, split into train/test:

File	Total	Consistent	Toy	Non-Toy
train_set1	9,443	**6,279**	395 (6.3%)	5,884
train_set2	9,443	**6,789**	399 (5.9%)	6,390
test_set1	4,113	**2,840**	70 (2.5%)	2,770
test_set2	4,113	**2,330**	66 (2.8%)	2,264

Sentence **consistent** iff described event possible in at least one valid microworld situation. Only consistent sentences are used for training and evaluation.
Repeat toy sent.s in training $4\times$ to give NNs best chance for basic event test.

Test-Only Described & Competing Scores

Experiment 1: 100 models total = 5 arch $\times$ 2 entity $\times$ 2 splits $\times$ 5 seeds



Left: described events. Right: competing events.
Hatched: no entity vectors. Solid: with entity vectors.
Dark red lines: ideal model performance.

Test-Only Advantage Scores

Recurrent models

		W	S	Cx	B
SRN	−ent	1.65	**1.16**	1.36	0.67
	+ent	**1.73**	1.12	**1.37**	0.74
GRU	−ent	1.72	1.13	1.30	0.65
	+ent	1.72	1.08	1.20	**0.78**
LSTM	−ent	1.68	1.14	1.28	0.67
	+ent	1.71	1.12	1.24	0.77

Attention models

		W	S	Cx	B
AbsPE	−ent	1.71	0.95	1.05	0.59
	+ent	1.72	1.02	1.10	0.68
RoPE	−ent	1.67	1.14	1.07	0.63
	+ent	1.70	1.12	1.05	0.69

W=Word, S=Sentence, Cx=Complex, B=Basic.

Word easy for all, Basic sharply separates gated RN from attention.

Main Findings

- **Difficulty has 2 dimensions**: test type & modifier complexity.
- **Gated recurrence matters on Basic Event**: GRU/LSTM (but not SRN) outperform attention on Basic-Event.
- **Entity vectors help most on Basic Event**: the largest Experiment 1 gains are GRU $+0.125$ and LSTM $+0.106$.
- **SRN is best on Complex Event**: the gating advantage is subtask/test-specific, not uniform.
- **More attention capacity does not close the gap**: H80 attention does not materially improve over H48.